Indian Institute of Science Education and Research Mohali

CHM203 v.2 - Chemical kinetics

End-Sem Examination, May 6, 2026

Total marks: 50; Time 2.30 pm – 5.00 pm

***(Important note:** Please write your Name and Roll No. on the Q-paper; please return the Q-Paper along with the answer sheet.)*

1. Prove that the most probable speed $\mathbf{c}^* = [\frac{2K_BT}{m}]^{1/2}$. Draw a tentative Maxwell-Boltzmann distribution curve for a nitrogen molecule and represent $c^*, \langle v \rangle$, and v_{rms} graphically in the same graph. **(3+2 Marks)**

2. A zero-order reaction is *50%* complete in *20 minutes*. How much time will it take for the same reaction to complete *75%*? (**5 Marks**)

3. Show graphically the dependence of reaction rate on substrate concentration for an enzyme that follows *Michaelis-Menten* kinetics and determine the constants and the turnover frequency (TOF). The initial velocities (v_0) and substrate concentrations (s_0) are given below:

v_0	s_0 (M^{-1})
0.3570	1.4286
0.3125	1.1111
0.2381	0.7692
0.1449	0.4545
0.1111	0.3030

(Graph=2 Marks; 1 mark for each constant and TOF); total: **5 Marks**)

4. It is assumed that the values of q_t, q_v and q_r for different molecules in a reaction *Atom + Linear molecule (diatomic)* $\longrightarrow$ *Nonlinear molecule* are equal.
Calculate the value of Arrhenius factor (A) at *298 K* for above reaction, if q_t =10^8, q_v=10 and q_r =1. (**5Marks**)

5. Estimate the thermal conductivity coefficient of an oxygen molecule at *600 K*, given that the diameter of the molecule is *346 pm*. {*Hint: mass of* O_2 *= 32 amu*} (**5 Marks**)

6. Find the number of oxygen molecules diffusing through a nitrogen molecule and crossing an area of *0.4* m^2 at *373K*. If the concentration gradient is *80 Torr/cm* and the diffusion coefficient for O_2-N_2 =1.8×10^{-5} m^2 sec^{-1}. {*Hint: 1 Torr = 0.0013158 atm*} (**5 Marks**)

7. (a) A drug molecule has a Brownian radius of *1 nm*. Calculate its diffusion coefficient at *298 K*, given a viscosity of *0.8 CPa*. (b) From the above diffusion coefficient, calculate the diffusion-limited rate for a pair of the same molecule with an average diameter of *2 nm*. {*Hint: 1 CPa = 0.001 Pa. sec.*} (**2.5+2.5 Marks**)

8. Prove (both schematically and mathematically) that the predicted Marcus rate equation for electron transfer reaction ***does not*** change linearly with free energy. (**2+3 Marks**)

9. Consider adsorption and desorption for a reaction: $A_2 + S + S \longrightarrow A\text{-}S + A\text{-}S$. If the surface coverage is θ show that θ changes with ½ power of A_2 concentration. (**5 Marks**)

10. What is Stern-Volmer quenching? Derive the equation to get the Stern-Volmer quenching constant (K_{SV}). Calculate the rate of quenching if K_{SV} is 10^2 M^{-1}, and the fluorescence lifetime is *10 ns*. (**5 Marks**)

Important universal constants:

1. $h = 6.626 * 10^{-34} Js$
2. $k_B = 1.38 * 10^{-23} J/K$
3. $c = 2.998 * 10^8 m/s$
4. $N_A = 6.023 * 10^{23}$ /mol